Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Srushti Chaudhari
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (EVEN)**	**Course Name**	**AIML Lab**
Semester	**6**	**Course Code**	**23CT1602**
Roll No	**23**	**Name of Student**	**Srushti Chaudhari**

Practical Number	1
Course Outcome	CO1: Apply Artificial Intelligence problem formulation techniques, intelligent agents, search algorithms, and logical inference to solve real-world problems. CO2: Implement and evaluate machine learning algorithms including classification, clustering, and dimensionality reduction using appropriate preprocessing and performance metrics. CO3: Analyze and compare ensemble learning models.
Aim	Smart Light Agent (Simple Reflex Agent) To design and implement a **Simple Reflex Agent** that simulates a **Smart Light System** which automatically turns the room light **ON or OFF** based on outside light availability and human presence.
Problem Definition	A smart light system should automatically control indoor lighting to save energy and improve convenience. The agent observes two percepts from the environment: **outside light availability** (sunlight available or not) and **human presence** (someone present in the room or not). Based on these percepts, the agent decides whether to turn the indoor light **ON** or **OFF**. The system follows simple reflex rules. If a person is present in the room and there is no outside light, the system should turn the light ON. In all other situations, the system should keep the light OFF. The goal is to simulate this decision-making process using a simple reflex agent model.
Theory (100 words)	A **Simple Reflex Agent** is a type of intelligent agent that selects actions based only on the **current percept**, ignoring past history. It works using **condition–action rules**, where a specific action is triggered when a particular condition is satisfied. In a **Smart Light System**, the agent observes two percepts: **human presence** in the room and **availability of outside light**. Based on these inputs, the agent decides whether the indoor light should be **ON or OFF**. If a person is present and there is no outside light, the agent turns the light ON; otherwise, it turns it OFF. This approach helps automate lighting and improves **energy efficiency**.

Procedure and Execution

(100 Words)

Algorithm:

1. Start the program.
2. Take input for **human presence** (Yes/No).
3. Take input for **outside light availability** (Yes/No).
4. Check the conditions using simple reflex rules:
   - If human is present AND outside light is not available → Turn **Light ON**.
   - Otherwise → Turn **Light OFF**.
5. Display the action taken by the smart light agent.
6. Stop the program.

Code:

```
def smart_light_agent(human_present, outside_light):

 if human_present == "yes" and outside_light == "no":
 action = "Light ON"
 else:
 action = "Light OFF"

 return action

 human_present = input("Is a human present in the room? (yes/no): ").lower()
 outside_light = input("Is outside light available? (yes/no): ").lower()

 result = smart_light_agent(human_present, outside_light)

 print("Smart Light Agent Decision:", result)
```

Output:

```
... Is a human present in the room? (yes/no): yes
    Is outside light available? (yes/no): no
    Smart Light Agent Decision: Light ON
```

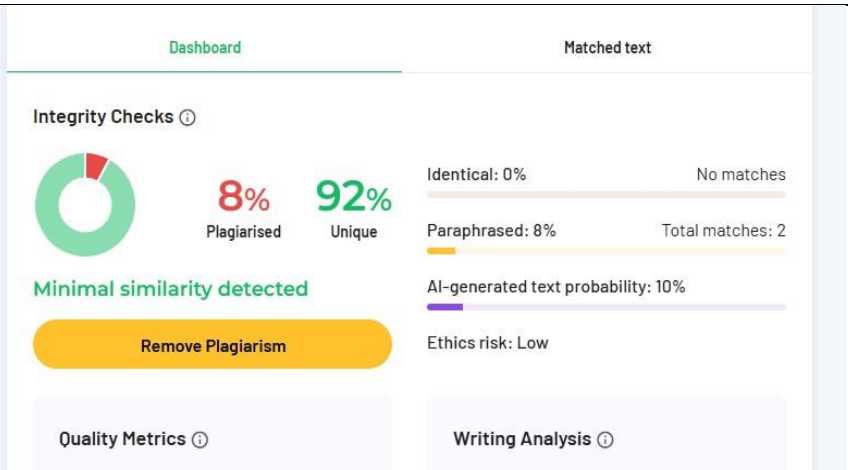
```
... Is a human present in the room? (yes/no): no
    Is outside light available? (yes/no): no
    Smart Light Agent Decision: Light OFF
```

```
... Is a human present in the room? (yes/no): yes
    Is outside light available? (yes/no): yes
    Smart Light Agent Decision: Light OFF
```

```
... Is a human present in the room? (yes/no): no
    Is outside light available? (yes/no): yes
    Smart Light Agent Decision: Light OFF
```

Output Analysis

The Smart Light Agent works based on simple reflex rules. It takes two percepts from the environment: human presence and outside light availability. If a person is present and there is no outside light, the agent turns the light ON. In all other situations, the agent turns the light OFF. This demonstrates how a simple reflex agent makes decisions based only on the current percept without considering past states. The system helps in automatic lighting control and energy efficiency.

Link of student Github profile where lab assignment has	https://github.com/SRchaudhari15/AIML-Practicals.git
Conclusion	The Smart Light Agent based on the **Simple Reflex Agent model** was successfully implemented. The agent makes decisions using simple condition–action rules based on two percepts: human presence and outside light availability. If a person is present and there is no outside light, the system turns the light ON; otherwise, it remains OFF. This demonstrates how simple reflex agents operate using current environmental inputs without considering previous states. The system helps automate lighting control and improves energy efficiency.
Plag Report (Similarity index < 12%)	
Date	29/12/25

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